Pollution

CHEMICALS FROM HUMAN ACTIVITIES AFFECT THE ECOSYSTEMS AND FUTURE OF THE EARTH.

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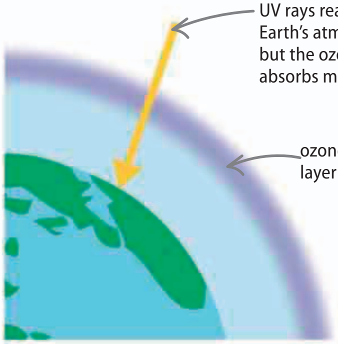
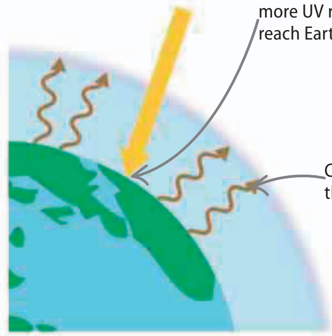
Pollution is anything that is added to the environment in amounts large enough to have a harmful effect. Sound, light, and heat can be pollution, but the most damaging pollutants are chemicals in Earth's soil, water, and air.

Ozone hole

Ozone is a type of oxygen in the atmosphere that blocks dangerous ultraviolet light (UV) coming from the Sun. Large amounts of chlorofluorocarbon (CFC) gases, used in aerosols and refrigerators and thought to be inactive, were released in the 1980s. The CFCs reacted with ozone, and, over the years, have depleted the ozone layer in places, especially above the North and South Poles. CFCs are now banned and the ozone holes are shrinking.



△ **Safe levels**
While some does hit the Earth's surface, the ozone layer, 25 km (15.5 miles) above Earth, deflects much of the harmful UV light back into space.



△ **High layer**
A chemical reaction between the CFCs and the ozone layer turns the latter into oxygen, which does not shield Earth from UV light, meaning more of it reaches the surface.

REAL WORLD

Global dimming

Burning fossil fuels releases carbon dioxide, which contributes to global warming. However, the soot released may also keep temperatures down in a process called global dimming. The tiny dark particles in the air reflect the Sun's light, reducing the amount that heats Earth's surface.

Greenhouse effect

The "greenhouse gases" of water vapor, carbon dioxide, and methane in the atmosphere stop heat being lost to space. Without this process, Earth's average surface temperature would be below freezing. However, human activities, such as burning fossil fuels and intensive farming, are increasing the amount of greenhouse gas. This greenhouse effect is gradually increasing Earth's surface temperature, resulting in more extreme weather, such as flooding and drought.

Venus, warm enough to melt lead, is the hottest planet, due to an **extreme** greenhouse effect.

▽ **Trapped heat**
Sunlight absorbed by the Earth's surface warms it up, and the surface sends out heat in the form of infrared radiation. Some infrared is absorbed in turn by the atmospheric greenhouse gases.